

MA374 – Lab 05 Report

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Question 1

We use the same three-asset data from Lab 04:

$$\mu = [0.10, 0.25, 0.15]$$

$$C = \begin{bmatrix} 0.005 & -0.01 & 0.004 \\ -0.01 & 0.04 & -0.002 \\ 0.004 & -0.002 & 0.023 \end{bmatrix}$$

The minimum variance portfolio problem (with no short selling) is:

$$\min_w w^T C w$$

subject to

$$w_1 + w_2 + w_3 = 1, \quad w_i \geq 0$$

Random portfolios were generated with non-negative weights and normalized so that the weights sum to one. For each portfolio we computed:

$$r = w^T \mu, \quad \sigma = \sqrt{w^T C w}$$

The feasible region is obtained by plotting all sampled portfolios. The minimum-variance curve is the lower envelope of this region, and the efficient frontier is the upper envelope. Pairwise two-asset minimum variance curves were also plotted by setting the third weight to zero.

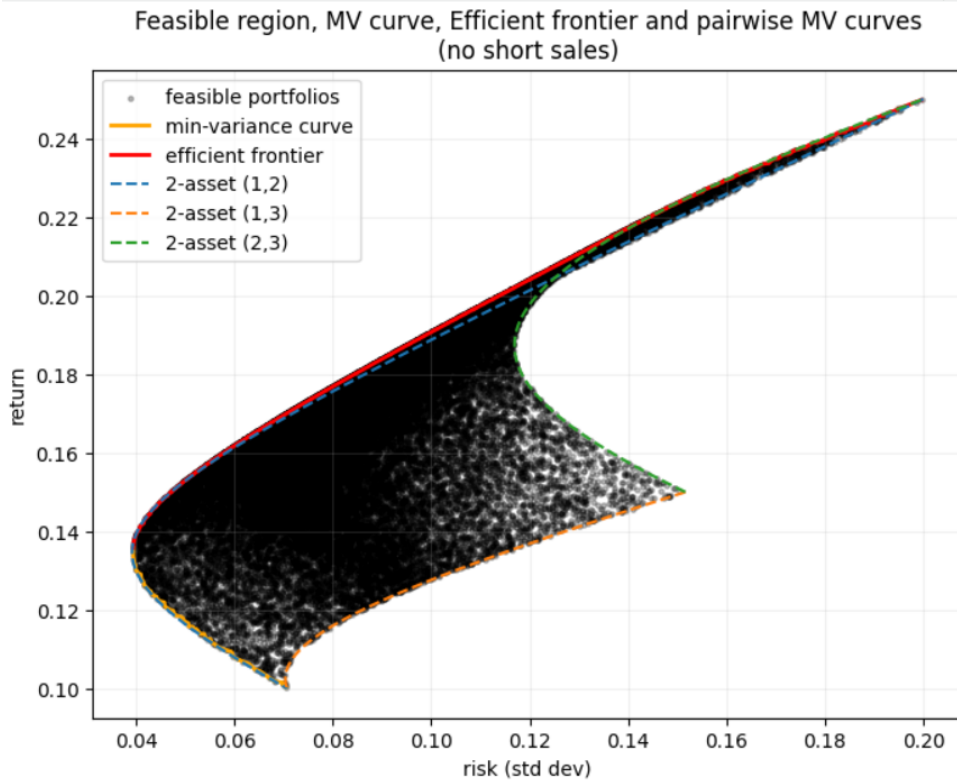


Figure 1: Feasible region, minimum variance curve, efficient frontier and pairwise two-asset curves (no short sales).

The weights of the three assets along the minimum-variance curve are shown below.

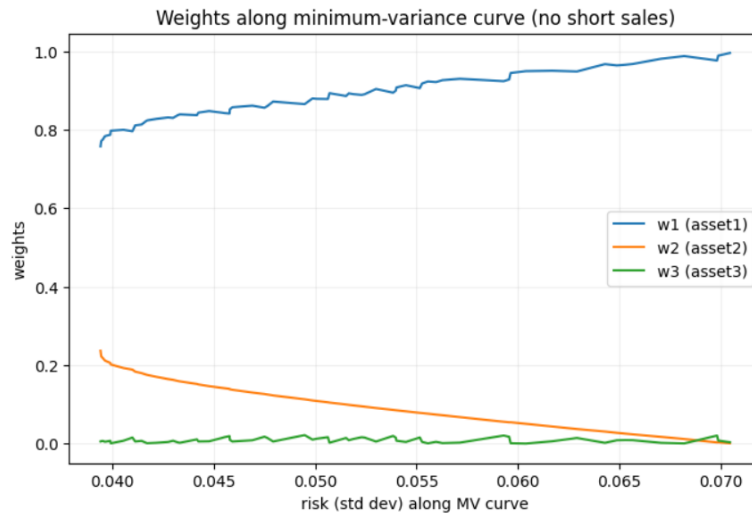


Figure 2: Asset weights along the minimum-variance curve.

Observation: Asset 1 dominates most of the minimum-variance curve, while asset 2 decreases gradually and asset 3 remains close to zero.

Question 2

Daily price data (2021–2025) was collected for NSE and BSE. Each dataset contains:

- Index (market portfolio)
- 10 index stocks
- 10 non-index stocks

Daily simple returns were computed as:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

The risk-free rate was taken as 5% annually and converted to daily:

$$r_f = \frac{0.05}{252}$$

For each stock, beta was estimated using:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

CAPM expected return:

$$E[R_i] = r_f + \beta_i(E[R_m] - r_f)$$

The Security Market Line (SML) was plotted along with all stocks.

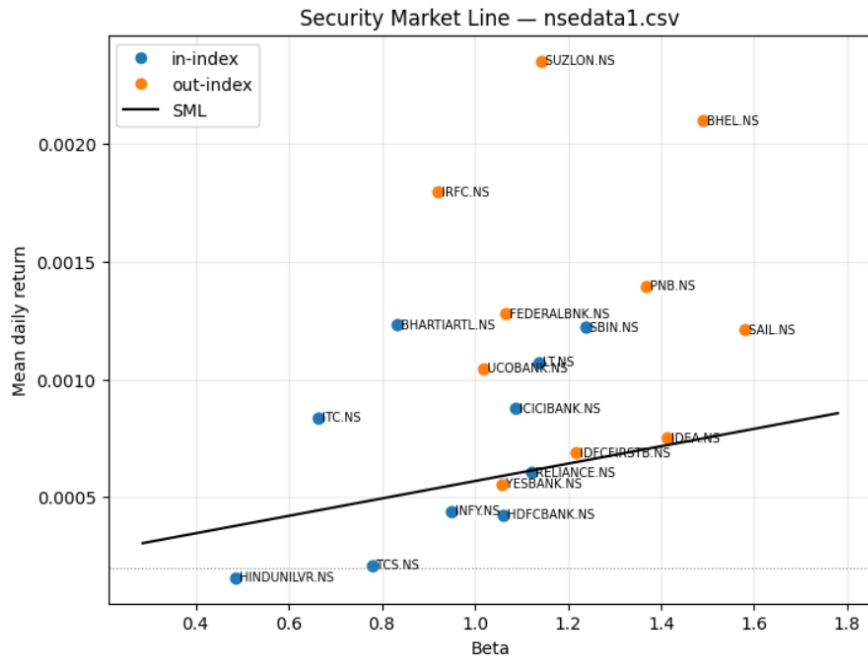


Figure 3: Security Market Line – NSE

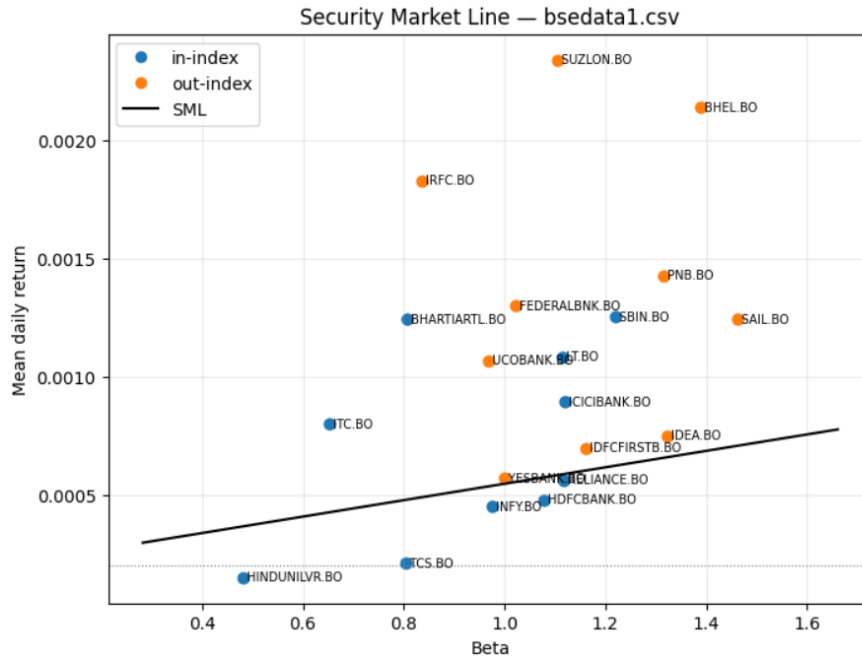


Figure 4: Security Market Line – BSE

Summary of Results

Average betas

- NSE: inside index = 0.936, outside index = 1.227
- BSE: inside index = 0.937, outside index = 1.158

Non-index stocks have higher average beta, meaning they carry higher systematic risk.

Several stocks such as SUZLON, BHEL, IRFC and PNB lie well above the SML, indicating higher realised returns than CAPM predicts during this period. Large index stocks (Reliance, TCS, INFY, HUL) lie close to or slightly below the SML.

Conclusion

In Question 1, the feasible region, minimum variance curve and efficient frontier were successfully obtained under no short selling. In Question 2, betas and CAPM returns were computed using NSE and BSE data and Security Market Lines were plotted.

Outside-index stocks show higher average beta in both markets. Some stocks outperform CAPM in this sample, while most large index stocks follow the SML closely.